

Cyclistic Bike-Share Analysis

Executive Report

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Project: Google Data Analytics Professional Certificate Capstone

Tools Used: PostgreSQL, SQL, pgAdmin 4, Tableau, GitHub

Date: July 2026

1. Executive Summary

Cyclistic is a bike-sharing company that aims to increase the number of annual memberships by converting casual riders into long-term subscribers. This project analyzes historical bike-share data to understand how annual members and casual riders use Cyclistic services differently. The analysis was conducted using PostgreSQL for data cleaning and exploration, and Tableau for visualization.

The findings show that annual members account for most rides and primarily use the service for regular transportation, while casual riders take longer rides and are more active during weekends and warmer months. Based on these behavioral differences, targeted marketing strategies are recommended to encourage casual riders to purchase annual memberships.

2. Business Problem

Cyclistic's marketing team believes that increasing annual memberships is more profitable than acquiring new casual riders. However, before creating marketing campaigns, the company needs to understand the behavioral differences between annual members and casual riders.

3. Stakeholders

- Cyclistic Executive Team
- Marketing Director
- Marketing Analytics Team

4. Business Task

Analyze historical bike-share trip data to identify behavioral differences between annual members and casual riders and provide data-driven recommendations that can help convert casual riders into annual members.

5. Dataset Description

The analysis used the Cyclistic (Divvy) Bike Share historical trip dataset provided through the Google Data Analytics Professional Certificate.

Dataset Period: April 2020 – March 2021

Records: 3,478,810

Columns: 16

The dataset contains information including:

- Ride ID
- Bike type
- Start and end time
- Start and end stations
- Geographic coordinates
- Rider category (Member or Casual)

6. Tools Used

Tool	Purpose
PostgreSQL	Data storage
SQL	Data cleaning and analysis
pgAdmin 4	Database management
Tableau	Dashboard creation
GitHub	Project documentation

7. Data Preparation

The following data preparation steps were completed:

- Imported twelve monthly CSV files into PostgreSQL.
- Combined all monthly records into a single database table.
- Verified data integrity by checking row counts.
- Checked for duplicate records (none found).
- Examined missing values in station names and geographic coordinates.
- Created additional variables including:
 - Ride Length
 - Day of Week
 - Month

8. Exploratory Data Analysis

Several analyses were performed to understand rider behavior.

Rider Distribution: Annual members completed **2,051,851** rides while casual riders completed **1,426,959** rides.

Bike Preferences: Docked bikes were the most frequently used bike type for both rider categories.

Average Ride Duration

Annual Members: 16 minutes

Casual Riders: 45 minutes

Casual riders generally spend almost three times longer on each trip.

Weekly Riding Trends

Annual members ride consistently throughout the week.

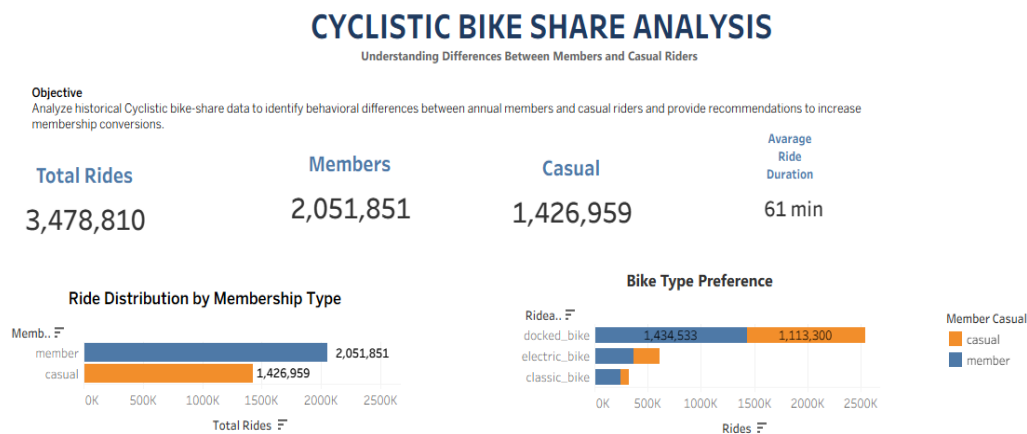
Casual riders show significantly higher activity during weekends.

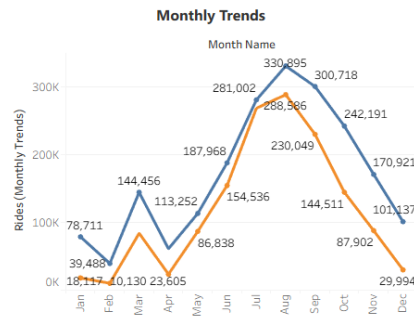
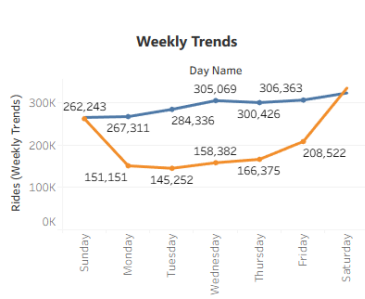
Monthly Riding Trends

Both rider categories experience increased activity during summer months.

Annual members maintain relatively stable usage throughout the year.

9. Dashboard Summary





KEY INSIGHTS

- Members generated 59% of all rides.
- Casual riders average nearly three times longer trips than members.
- Members primarily ride during weekdays, suggesting commuting behavior..
- Casual ridership peaks on weekends, especially Saturdays.
- Classic bikes are the most popular choice for both member types.

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The Tableau dashboard summarizes rider behavior using key performance indicators, rider distribution, bike preferences, weekly trends, monthly trends, and business insights. Interactive filters enable users to explore the data efficiently.

10. Key Findings

1. Annual members completed more rides than casual riders.
2. Casual riders have much longer average ride durations.
3. Casual rider activity increases significantly during weekends.
4. Bike usage peaks during summer months.
5. Docked bikes remain the preferred bike type for both groups.

11. Business Recommendations

Based on the analysis, the following recommendations are proposed:

- Launch weekend membership promotions targeting casual riders.
- Introduce discounted annual memberships during peak riding seasons.
- Develop marketing campaigns highlighting the financial savings of annual memberships.
- Provide loyalty rewards for frequent casual riders.
- Offer commuter-focused membership packages for riders with regular travel patterns.

12. Conclusion

This analysis demonstrates that annual members and casual riders exhibit different riding behaviors. Casual riders primarily use the service for leisure activities, while annual members rely on the system for more regular transportation. Understanding these differences enables Cyclistic to design targeted marketing strategies that encourage casual riders to transition into annual members, ultimately increasing long-term profitability.

13. References

Google Data Analytics Professional Certificate

Cyclistic (Divvy) Bike Share Historical Dataset

Tableau

PostgreSQL Documentation